

# Hanson H. Nguyen

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## Education

### University of California Berkeley, Berkeley, CA

Aug 2026 - Present

- Ph.D in Electrical Engineering and Computer Science
- Awards: NSF Graduate Research Fellowship Program

GPA: TBD/4.00

### Arizona State University, Tempe, AZ

Aug 2022 - Dec 2025

- B.S.E, Electrical Engineering, *Minor: Mathematics; Barrett, The Honors College*
- Awards: Goldwater Scholar, MIT Summer Research Program, New American University - National Merit, Dean's list (all academic semesters), Moeur Award, Impact Award, Electrical Engineering Outstanding Graduate, Regalia Award

GPA: 4.00/4.00

## Skills/Interests

### Computational & Design Tools

- Python and MATLAB for scientific computing and data analysis; Bayesian inference and MCMC methods; GNU/Linux environments; SystemVerilog for FPGA RTL design; SPICE circuit simulation; Cadence and Silvaco TCAD; KiCad and Fusion 360 for electronic and mechanical design; GitLab CI/CD for version control and collaborative development; digital and hardware communication protocols (I<sup>2</sup>C, UART, SPI, ETH);

### Experimental Hardware & Systems

- FPGA-based control and signal-processing systems; microcontrollers and embedded platforms; cryogenic systems and cryogenic electronics; fiber-optic and electro-optic hardware; perovskite solar cell fabrication; optoelectronic device characterization and testing

## Project/Research Experience

### Argonne Medium Energy Physics Group

Jan 2026 - Present

#### Seasonal Intern

- Developed custom RTL platform for high speed data acquisition on Zynq Ultrascale+ MPSoC FPGA development board for superconducting nanowire detector readouts in Vivado design suite.
- Developing experimental set up and characterizing radiation hardness and detection efficiency of superconducting nanowire detectors for alpha particle detection

### Purdue Quantum Networking Lab

Oct 2023 - Dec 2025

#### Undergraduate Researcher

- Designed unorthodox parallel preconditioned Crank-Nicolson (pCN) Metropolis Hastings and parallel sequential Monte Carlo (SMC) algorithms for quantum state tomography (QST) on up to 4-qubits on superconducting qubits using IBM quantum computers with Qiskit
- Developed threshold quantum state tomography scheme for frequency-bin photonic quantum information using electro-optic modulators and fiber optics

### MIT Quantum Nanostructures and Nanofabrication Lab

June 2024 - Aug 2025

#### MIT Summer Research Program Intern

- Built an electrothermal compact model of superconducting nanowire single photon detectors to simulate hotspot dynamics and recovery implemented in SPICE.
- Performed cryogenic photon-correlation measurements to characterize SNSPD reset times and implemented post-processing filters and event detection to improve SNR and extract correlations.

### ASU Renewable Energy Materials and Devices Lab

Feb 2023 - Dec 2023

#### Undergraduate Researcher, funded by FURI and GCSP, sponsored by GORE

- Investigated additive engineering, ion migration, and accelerated aging of all-inorganic cesium-based perovskites
- Designed and conducted experiments to develop thermally and mechanically stable thin films via processing techniques such as spin coating, metal evaporation, and blade coating.
- Characterized films' optical, mechanical, and electrical properties via photoluminescence, UV-Vis Spectroscopy, x-ray diffraction, microscopy, and more

### RET SITE: Collaborative Research: Sonoran Photovoltaic Laboratory

May 2023 - May 2024

#### Lead Program Student Mentor / Research Assistant

- Developed curriculum on photovoltaics, agrivoltaics, and electrical engineering for citizen science research
- Engineered compact LiFePO<sub>4</sub> battery solar energy control systems for local teachers to engage in classroom research

## Publications/Presentations

### Peer-Reviewed Publications

- **Nguyen H**, Su CY, Maragnano D, Liscidini M, McKinney JD, Lu HH, Lukens JM. 2026. Measurement-efficient tomography of high-dimensional frequency-bin quantum states (In preparation)

- **Nguyen H**, Simon A, Foster R, Berggren K. 2025. Modeling electrothermal feedback of superconducting nanowire single photon detectors in SPICE. *IEEE Transactions on Applied Superconductivity (Published)*
- **Nguyen H**, Law KJH, Lukens JM. 2025. Unorthodox parallelization for Bayesian quantum state estimation. *New Journal of Physics. (Published)*
- **Nguyen H**, Penukula S, Mahaffey M, Rolston N. 2024. All inorganic CsPbI<sub>3</sub> perovskite solar cells with reduced mobile ion concentration and film stress. *MRS Communications. (Published)*
- **Nguyen H**, Su CY, Maragnano D, Liscidini M, McKinney JD, Lu HH, Lukens JM. 2026. Measurement-efficient tomography of high-dimensional frequency-bin quantum states (In preparation)
- **Nguyen H**, Liang X, Lohani S, Clark BR, Kirby BT, Law KJ, Peters NA, Searles TA, Lukens JM. 2025. Parallel Bayesian quantum state estimation with strong scaling. (In preparation)

#### Conference Abstracts

- **Nguyen H**, Su C, Maragnano D, Liscidini M, McKinney JD, Lu HH, Lukens JM. Measurement-efficient tomography of high-dimensional frequency-bin quantum states. *Quantum 2.0, June 2026*.
- **Nguyen H**, Law KJH, Lukens JM. Unorthodox parallelization for Bayesian quantum state estimation. *CLEO, May 2025*.
- Jordan M, Bendok S, **Nguyen H**, Zuiker S, Potter-Simmons K, Meza-Torres C, Barron-Gafford G, Honsberg C, Boyer K, Jongewaard R, Loya B, Ankrum J. Sonoran Desert Photovoltaics Laboratory and Growing Green: A Networked Regional Approach to Agrivoltaics Citizen Science. *IEEE 52nd PVSC, June 2024*.

#### Presentations

- **Nguyen H**, Su C, Maragnano D, Liscidini M, McKinney JD, Lu HH, Lukens JM. Measurement-efficient tomography of high-dimensional frequency-bin quantum states. *Quantum 2.0, June 2026*.
- **Nguyen H**, Law KJH, Lukens JM. Unorthodox parallelization for Bayesian quantum state estimation. *CLEO, May 2025*.
- **Nguyen H**, Simon A, Berggren K. Thermal and electronic reset times of superconducting nanowire single photon detectors. *MIT Summer Research Program Poster Showcase, August 2025*
- **Nguyen H**, Simon A, Berggren K. Electrothermal Model of Superconducting Nanowire Single Photon Detectors. *MIT Summer Research Program Poster Showcase, August 2024*
- **Nguyen H**, Simon A, Berggren K. Electrothermal Model of Superconducting Nanowire Single Photon Detectors. *Poster session presented at: MIT Digital Poster Showcase; November 2024*
- **Nguyen H**, Rolston N. Scaling of All-Inorganic Perovskite Solar Cells through Improved Thermomechanical and Optoelectrical Stability. *Fulton Forge Student Research Expo; November 2023 (Outstanding Undergraduate Award)*

#### Leadership/Involvement

- |                                                      |                            |
|------------------------------------------------------|----------------------------|
| - Electrical Engineering Student Ambassador          | <i>May 2024 - Present</i>  |
| - ASU IEEE Student Branch (Member)                   | <i>Fall 2023 - Present</i> |
| - Quantum Club Founder (Founder)                     | <i>Feb 2024 - May 2025</i> |
| - Quantum Mechanics Course Grader                    | <i>Aug 2024 - Dec 2024</i> |
| - Digital Design and Logic Course Teaching Assistant | <i>Aug 2023 - Dec 2024</i> |
| - Quantum Mechanics Course Teaching Assistant        | <i>Jan 2024 - May 2024</i> |
| - Engineering Projects in Community Service          | <i>Aug 2022 - May 2023</i> |

#### Relevant Coursework

##### Physics

- General Physics I, II, and III, Quantum Mechanics for Quantum Information Science, Introduction to Quantum Computing and Information (**graduate**), Fundamentals of Electromagnetics, Properties Electronic Material, Fundamentals of Solid State Devices, Introduction to Solid State Electronics (**graduate**), Semiconductor Device Modeling (**graduate**)

##### Engineering

- Circuits I, II, and III (Analog and Digital Circuits), VLSI Device Circuit Co-design (**graduate**), Engineering Electromagnetics, Hardware Design Language/Programmable Logic, Random Signal Analysis, Digital Logic and Design, Signals and Systems I, Computer Organization/Assembly Language Programming

##### Mathematics

- Statistics, Differential Equations, Multivariable Calculus, Linear Algebra, Mathematical Structures, Applied Complex Analysis

##### Computer Science

- Applied Computational Methods, Principles of Programming, Python Programming for Physicists, Mathematical Tools for Data Science